

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		<u>Enzyme substrate complex</u>	1			1		
		(ii)		Lock and key	1			1		
	(b)	(i)		A. Volume of {oxygen/gas} increases up until {38cm ³ /25 minutes} (1)	1					1
				B. (because) {hydrogen peroxide/ substrate} is being {broken down/ used up} (1)			1			
				C. Volume of {oxygen/ gas} then {remains constant/ stops increasing/ stays the same/ plateaus}/ no gas is produced after {38cm ³ /25 minutes} (1)	1			4		1
		(ii)		D. (because) <u>all</u> the {hydrogen peroxide/ substrate} has been {broken down/ used up}/ <u>no</u> substrate left (1) IGNORE ref. denature at 25/30min			1			
				0.4 = 2 marks If incorrect award 1 mark for (7-5)/5 or 2/5		2		2	2	1
		(iii)		- Copper sulfate has bound to (some of the) {catalase/enzyme} / {fewer/some} enzymes are working/ (some) enzymes are not working} (1) - Shape of those active sites has changed / {active site/catalase/enzyme} is denatured (1) - Enzyme substrate complex cannot form / {hydrogen peroxide/substrate} cannot {fit into/ bind with} the {active site/enzyme/catalase} / {No/ fewer} successful collisions between {active site/ catalase/ enzyme} and {hydrogen peroxide/ substrate} (1)		3		3		